

Fatigue-resistant metal-film-based flexible conductors with a coherent gradient nanolayered architecture

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Metal-film-based conductors are an important element of flexible electronic devices. However, they typically suffer from fatigue damage and electrical degradation under cyclic deformation, which can limit practical use. Here we report fatigue-resistant metal films with a coherent gradient nanolayered architecture. The architecture consists of alternating stacked layers of silver and aluminium, with silver layers that become progressively thinner and finer grained. Initial crack nucleation is delayed by a combination of heterodeformation-induced strengthening, controlled grain coarsening in the silver layer and mitigation of interface stress concentrations. The moderate interface adhesion between silver and aluminium, and the multiaxial stress state induced by the gradient structure, also promote interface delamination and crack deflection, which suppresses fatigue-crack propagation. Our coherent gradient nanolayered silver/aluminium films exhibit a conductivity of over 10^7 S m^{-1} and relatively little conductivity change in both high-cycle, low-stress regimes (10^7 cycles at 0.7% strain) and low-cycle, high-stress regimes (10^5 cycles at 5% strain).

Flexible conductive materials are of use in applications such as deformable displays^{1–4}, wearable electronics^{5–8}, implantable bioelectronics^{9–12} and soft robotics^{13–15}. Metal films on compliant substrates can exhibit intrinsically high conductivity ($>10^7 \text{ S m}^{-1}$) and are easy to integrate with conventional microelectronics fabrication technology¹⁶. However, they typically suffer from poor mechanical flexibility (low tensile ductility and short fatigue lifetime) due to the effect of grain boundaries on dislocation activities. Various approaches have been developed to improve the stretchability of metal films, including adding interlayers^{17,18}, enhancing interfacial adhesion^{19–21}, designing layered architectural composites^{22,23} and using strain-relief structures (for example, in-plane serpentine²⁴ or out-of-plane buckling structures^{25,26}).

There has however been less focus on designing conductive materials that resist cyclic fatigue, which is a common cause of catastrophic failure of microelectronic systems in real-world applications^{27,28}. Fatigue failure can occur at strain levels that are low ($<2\%$) compared

with the static fracture strain under monotonic loading, leading to conductivity reduction and even complete electrical disconnection. Recently, a two-dimensional interlayer strategy was shown to improve the stretchability and low-cycle fatigue resistance of metal-film-based conductors by changing the in-plane fracture modes from straight paths to deflected/tortuous paths²⁹. However, achieving cycling stability of micro-/nanoscale metal-film-based conductors in both low- and high-cycle fatigue regimes remains challenging.

Fatigue damage originates from the progressive accumulation of localized permanent structural changes due to the repeated application of stresses or strains^{30,31}. Accumulation of plastic strain can create local microstructural instabilities (for example, grain coarsening and shear-banding formation), and form mechanical heterogeneities between deformed and pristine regions that ultimately lead to fatigue-crack nucleation. Once nucleated, a fatigue crack tends to propagate rapidly along the grain boundaries in nanocrystalline

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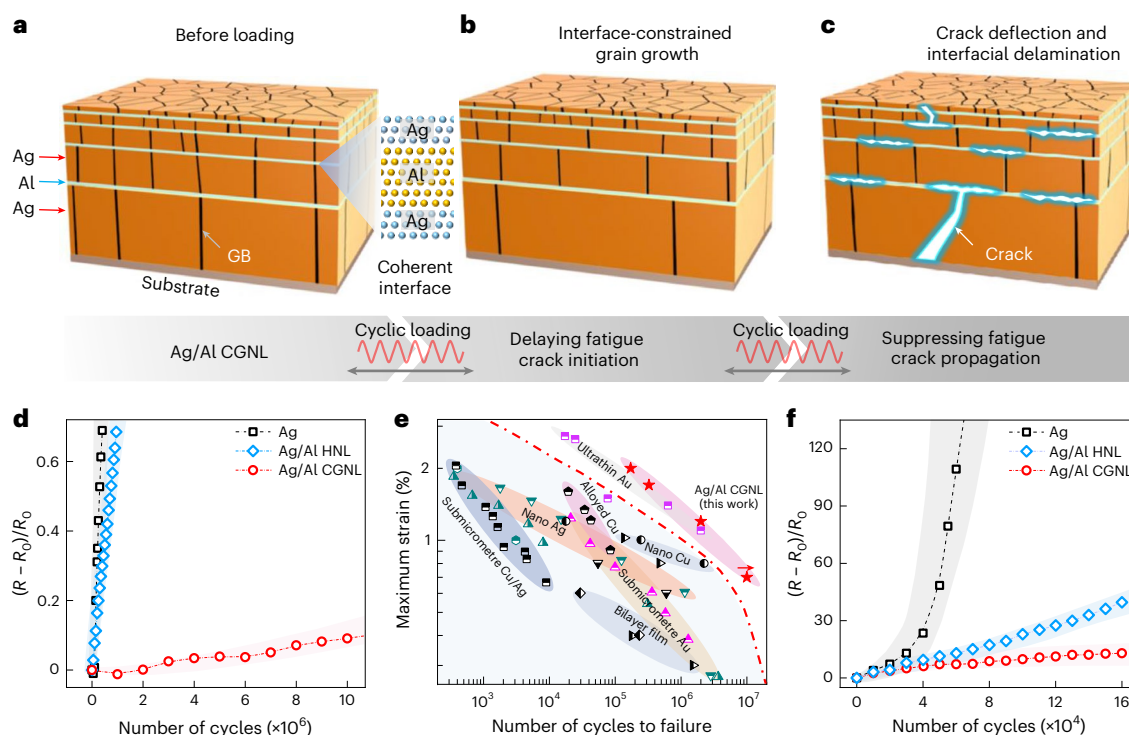


Fig. 1 | Fatigue-resistant flexible metal-film conductors enabled by the CGNL architectures. **a**, Schematic of the Ag/Al CGNL film featuring coherent interfaces, heterogeneous nanolayers and layer-thickness (grain-size) gradient. GB, grain boundary. **b**, Schematic of microstructure evolution during cyclic loading, illustrating controlled grain coarsening. **c**, Schematic of damage formation during cyclic loading, illustrating interface delamination and crack deflection. **d**, Relative resistance change as a function of fatigue cycles at a

maximum strain of 0.7% for Ag, Ag/Al HNL and Ag/Al CGNL films. R_0 is the initial electrical resistance and R is the real-time electrical resistance under loading. **e**, Comparison of high-cycle fatigue data with other reported metal conductors in the literature^{33–38}. **f**, Relative resistance change as a function of fatigue cycles at a maximum strain of 5% for Ag, Ag/Al HNL and Ag/Al CGNL films. The shaded regions in **d**, **f** are error bars corresponding to their coloured lines ($n = 3$ independent samples).

metals due to the limited plasticity and lack of crack path tortuosity. Therefore, delaying initial fatigue-crack nucleation and suppressing its propagation can offer a potential route to boost fatigue tolerance and increase the lifespan of metal-film-based conductors.

In this Article, we report a coherent gradient nanolayered (CGNL) architecture that can be used to create metal films with high fatigue resistance via a crack control strategy. The CGNL consists of alternating stacked silver/aluminium films where there is a gradient in the silver layer thickness (Fig. 1a). We select aluminium interlayers in a silver matrix due to the small mismatch in shear modulus (~30 GPa for Ag and ~26 GPa for Al) and lattice parameter (~0.409 nm for Ag and ~0.405 nm for Al), as well as their moderate interface adhesion energy³² (Supplementary Fig. 1 and Supplementary Table 1). High resistance to fatigue-crack initiation results from controlled grain coarsening in the silver layers, phase boundary strengthening and elimination of interface stress concentrations (Fig. 1b). Furthermore, maintaining a structural gradient and moderate interface adhesion promotes interface delamination and crack deflection, which inhibits fatigue-crack propagation (Fig. 1c). The resulting Ag/Al CGNL films exhibit an electrical conductivity of $3.1 \times 10^7 \text{ S m}^{-1}$ and high fatigue resistance, maintaining a high conductivity over 10^7 cycles at strains ranging from 0.7% to 2%, and over 10^5 cycles at a maximum strain of 5%. We show that our Ag/Al CGNL films can be used as fatigue-resistant electrodes and interconnects, and in bioelectronic interfaces.

Microstructural characteristics and fatigue performance

We performed real-time electrical resistance measurements to quantitatively characterize the fatigue properties of the CGNL films at maximum strains ranging from 0.7% to 5%. The fatigue life in the high-cycle

regime is defined as the critical number of cycles corresponding to 10% resistance increase, at which point out-of-plane cracks dominate throughout (Supplementary Fig. 2)³³. Figure 1d shows the electro-mechanical responses at a maximum strain of 0.7%. To emphasize the improvement in fatigue resistance with the CGNL architectures, single-layered Ag films (Supplementary Fig. 3a–c) and homogeneous nanolayered (HNL) Ag/Al films (Supplementary Fig. 3d–f) are presented for comparison. The CGNL film exhibits only a slight increase in electrical resistance after 10^7 cycles, and still preserves a conductivity that is orders of magnitude higher than those of Ag films and HNL films. Moreover, the CGNL film demonstrates superior fatigue resistance compared with conventional multilayered architectures (Supplementary Note 1 and Supplementary Figs. 4–8). We compared the fatigue lifetime of our CGNL film with those of other metal films^{34–36}, nanotwinned films^{33,37}, metal bilayers³⁸ and metal alloy films³⁶ (Fig. 1e). Our CGNL film achieves a fatigue resistance within wide maximum strains ranging from 0.7% to 2%, outperforming other metal-film-based conductors. The fatigue lifetime monotonically decreases as the maximum strain increases (Supplementary Fig. 9). Top-view scanning electron microscopy (SEM) observations confirm the superior fatigue resistance of the CGNL film compared with Ag and HNL films (Supplementary Fig. 10). No extrusions or cracks were observed in the CGNL films after 10^6 cycles, in contrast to the Ag films, with intensive extrusions/microcracks, and the HNL Ag/Al films, with a few extrusions/microcracks.

For low-cycle fatigue under large maximum strains (such as 5%), out-of-plane through-thickness cracks formed in the initial few hundred cycles (Supplementary Fig. 11), followed by propagation along the in-plane direction on further cyclic loading. The resistance change following the initial few hundred cycles therefore reflects the

resilience to in-plane propagation of fatigue cracks. When cycled at a maximum strain of 5%, the resistance of the CGNL film monotonically increases but preserves a small relative change of ~9 after 10^5 cycles, which is orders of magnitude lower than that of the Ag film (Fig. 1f). Top-view SEM images reveal substantially shorter cracks and lower crack densities in the CGNL film compared with Ag and Ag/Al HNL films (Supplementary Fig. 12), indicating a high resistance to in-plane propagation of fatigue cracks. In addition to cyclic tension, we characterized the electromechanical behaviour under cyclic bending. Fatigue damage tends to be localized at the upper surfaces of the metal films (Supplementary Fig. 13), where the in-plane tensile strain imposed via bending is a maximum due to the smallest curvature of radius. Our CGNL film exhibits cycle-insensitive electrical conductivity ($-3.1 \times 10^7 \text{ S m}^{-1}$) during cyclic bending at a maximum strain of 0.7%, indicative of an excellent bending fatigue property (Supplementary Fig. 14). By contrast, the electrical resistance of Ag and Ag/Al HNL films increased by ~115% and ~28% after 10^6 cycles, respectively. To ensure measurement reliability, three independent specimens of each film type were characterized under identical test conditions. The electromechanical response curves show a high degree of overlap (Supplementary Fig. 15), with a coefficient of variation below 10% (Supplementary Tables 2 and 3), demonstrating excellent reproducibility.

Fatigue resistance mechanisms

To understand the fatigue tolerance of the Ag/Al CGNL film, the microstructural features were characterized in detail. Atomic force microscopy images show very smooth surfaces with a surface roughness (root mean square) less than 5 nm (Supplementary Fig. 16). Cross-sectional transmission electron microscopy (TEM) imaging reveals the gradient nanolayered architecture composed of alternating stacked Ag layers and Al layers (Fig. 2a). The gradient of Ag layer thickness also generates a grain size gradient, because the out-of-plane grain size is of the order of the layer thickness (Fig. 2b) and the in-plane grain size increases with increasing layer thickness (Fig. 2c). Precession electron diffraction (PED) reveals that continuous and epitaxial columnar structures grow across multiple Ag and Al layers in the Ag/Al CGNL film (Fig. 2d). Pole figures reveal a preferred (111) out-of-plane orientation along the film growth direction (Fig. 2d). Transmission Kikuchi diffraction (TKD) orientation maps reveal preferred (111) out-of-plane orientations in Ag and Ag/Al HNL films (Supplementary Fig. 17).

We performed aberration-corrected high-angle annular dark-field transmission electron microscopy (HAADF-STEM) to characterize the atomic structures of Ag/Al interfaces (Fig. 2e–g). A clearly defined Al interlayer was sandwiched between the Ag layers. No substantial elemental interdiffusion occurs between the Al interlayer and the Ag layer, as evidenced by energy-dispersive X-ray spectroscopy (EDS) (Fig. 2e). High-resolution HAADF-STEM images taken with the incident electron beam aligned along the $[011]_{\text{Ag}}$ (Fig. 2f) and $[012]_{\text{Ag}}$ axes (Fig. 2g) reveal a fully coherent interface between the Al layer and Ag layer. Fast Fourier transform patterns (Fig. 2f(i),g(i)) have the same arrangements of spots corresponding to Ag and Al phases. The absence of additional spots indicates that no intermetallic phase formation occurs during the deposition process, also verified by X-ray diffraction (XRD) results (Supplementary Fig. 18). Inverse fast Fourier transform patterns (Fig. 2f(ii),g(ii)) further confirm the epitaxially coherent interface in a cube-on-cube orientation relationship, without misfit dislocations and/or atomic steps (Fig. 2h). Our CGNL film architecture exhibits fully coherent Ag/Al interfaces, heterogeneous nanolayers and a layer thickness/grain size gradient that is fundamentally distinctive to conventional metal films.

The high fatigue tolerance of the CGNL film stems from simultaneously improved resistance to initiation and to propagation of fatigue cracks. To illuminate the underlying mechanisms, we first characterized the microstructural evolution of the CGNL specimens before

crack initiation. Cross-sectional TEM images reveal intact nanolayered structures without interface voids and surface extrusions even after 10^6 fatigue cycles (Fig. 3a). Grain growth occurs in the bottom three Ag layers along the direction parallel to the Ag/Al interface, due to the grain-boundary migration mechanism^{39,40}. However, the Ag grains remain almost constant in the two topmost Ag layers (Supplementary Fig. 19), which is attributed to the deformation mechanism transition from full dislocation activities to the emission of partial dislocations (Supplementary Note 2). The stable Ag/Al interfaces can effectively suppress through-thickness grain growth while permitting lateral coarsening during cyclic loading.

We attribute the high fatigue initiation resistance to (1) heterodeformation-induced (HDI) strengthening, (2) controlled grain coarsening and (3) coherent Ag/Al interfaces. First, in gradient nanostructured metals, HDI stress emerges from both the self-stress and mutual interactions of geometrically necessary dislocations during plastic deformation⁴¹. Because different parts of the CGNL film deform unevenly, local strain gradients promote the formation of extra (geometrically necessary) dislocations that strengthen the material without causing brittleness. This hardening mechanism helps distribute deformation more uniformly, thereby maintaining good deformability. Thus, HDI strengthening enhances yield strength and concurrent strain hardening preserves deformability through the structural-gradient driven accumulation and interaction of dislocations. The HDI strengthening reduces the plastic strain amplitude compared with homogeneous counterparts, thereby enhancing the resistance to crack initiation.

Second, fatigue cracks preferentially initiate in abnormally grown grains where strain is localized⁴². Abnormal grain coarsening frequently occurs in conventional nanocrystalline metal films (abnormal grain coarsening in Ag films is shown in Supplementary Figs. 20 and 21), but the controlled grain coarsening in our CGNL film greatly improves fatigue-crack initiation resistance. This enhancement arises from two key mechanisms: suppression of through-thickness grain growth and promotion of lateral coarsening. Moreover, the surface Ag layer will suppress surface roughening (Supplementary Fig. 22) due to the constraints of the grain boundary and heterophase boundary on dislocation activities, thus postponing crack initiation.

Finally, the coherent Ag/Al interfaces can effectively alleviate stress concentration at interfaces and thereby delay the formation of interface voids/cracks. The coherent cube-on-cube interfaces with continuous crystal lattices facilitate transmission of dislocations across the interfaces under moderate stress and thus mitigate the stress concentration induced by dislocation accumulation. The moderate resistance of coherent Ag/Al interfaces to dislocation transmission mainly originates from the chemical interaction due to the stacking-fault energy difference^{43,44} (Supplementary Note 3). TEM observations on fatigued CGNL films show that a large number of dislocations are active within the Ag layer and in the vicinity of the Ag/Al interface (Fig. 3c), implying good deformation compatibility of the coherent interface. PED and corresponding kernel average misorientation images (Fig. 3d,e) also indicate dislocation activities within and across the Ag/Al interfaces during cyclic loading. Molecular dynamics simulations demonstrate that coherent interfaces (for example, Ag/Al) exhibit superior stability and lower strain concentration compared with semicoherent interfaces (for example, Ag/Mo) under cyclic loading (Supplementary Note 4 and Supplementary Figs. 23–26).

Fatigue-crack propagation resistance mechanisms

With progressive cyclic damage accumulation, void-initiated fatigue cracks preferentially nucleate at the film surface, Ag/Al interface and film–polyimide (PI) interface (Fig. 3b,f–h). We attribute the high crack propagation resistance to (1) crack-tip blunting, (2) interface shielding and (3) crack deflection. First, the fatigue cracks nucleated at the surface layer can be arrested by plastic-deformation-induced crack-tip blunting as the crack propagates towards adjacent coarser-grained

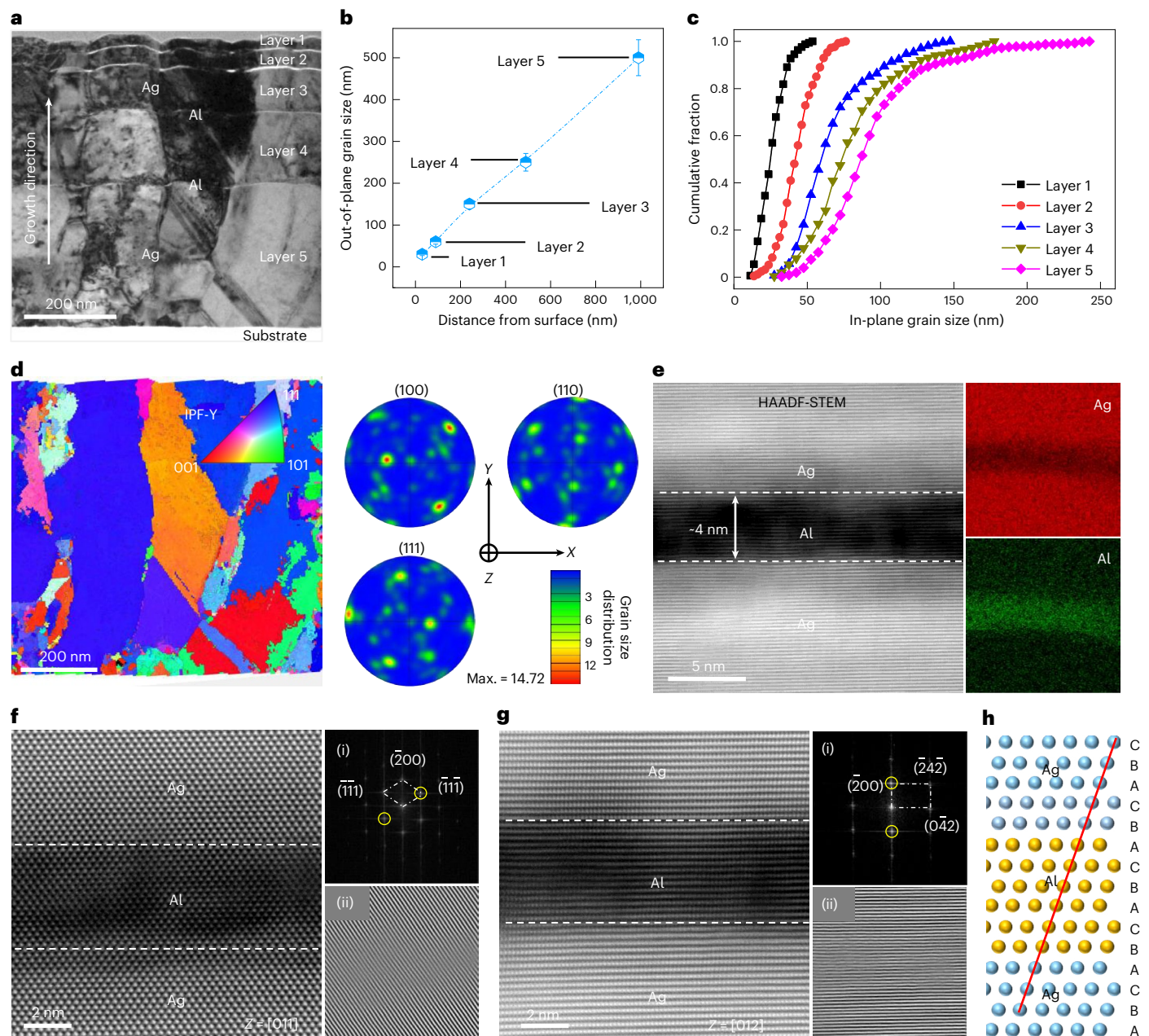


Fig. 2 | Microstructural characterization of the as-fabricated Ag/Al CGNL film.

a, Cross-sectional TEM image of the Ag/Al CGNL film. **b**, Variation of average out-of-plane grain size with layer thickness. The error bars are the s.d. of the mean ($n = 200$ grains). **c**, Cumulative distribution function of in-plane grain size. **d**, PED of Ag/Al CGNL film mapped with crystal orientation contrast (Y axis) and corresponding pole figures. IPF- Y refers to the inverse pole figure along the Y direction (film growth direction). **e**, HAADF-STEM image of the Ag/Al interface

and corresponding EDS colour mapping. **f, g**, Atomic-resolution HAADF-STEM images taken from the $\langle 011 \rangle$ (**f**) and $\langle 012 \rangle$ zone axes (**g**), showing the coherent Ag/Al interface with cube-on-cube orientation relationship. Insets: fast Fourier transform (**f**(i), **g**(i)) and inverse fast Fourier transform patterns (**f**(ii), **g**(ii)) of the Ag/Al interface, oriented to the $\langle 011 \rangle$ (**f**) and $\langle 012 \rangle$ zone axes (**g**), respectively. Z is the zone axis. **h**, Schematic of the coherent Ag/Al interface across which slip systems are continuous.

layers (Fig. 3f–h). The stress intensity factor, which describes the driving force behind the crack propagation, increases with increasing external stress and crack length⁴⁵. Owing to the thin surface Ag layer, a tiny surface crack would be easily blocked by the thicker Ag layer via plastic blunting (Fig. 3h).

Second, the fatigue cracks nucleated at the Ag/Al interface or film–PI interface tend to be arrested (Fig. 3g,h) as they advance towards the interface from the plastically soft Ag side, which we call the interface shielding effect. Material inhomogeneity, associated with gradients in yield stress at Ag/Al interfaces (Supplementary Fig. 27), exists in Ag/Al CGNL films. Therefore, when a crack approaches the

interface from the Ag layer towards the Al layer, the interaction of the crack-tip plastic zone with the interface can reduce the local stress intensity at the crack tip and the crack driving force^{46,47}, thereby impeding crack propagation.

Third, interface delamination and crack deflection at the Ag/Al interface appears in CGNL films (Fig. 3b,d–h), which reduces the mode I crack driving force and notably enhances the total crack length, maximizing the resistance against fatigue-crack propagation. A crack impinging on an Ag/Al interface may arrest or advance by either penetrating the interface or deflecting into the interface. The competition between deflection and penetration at the interface

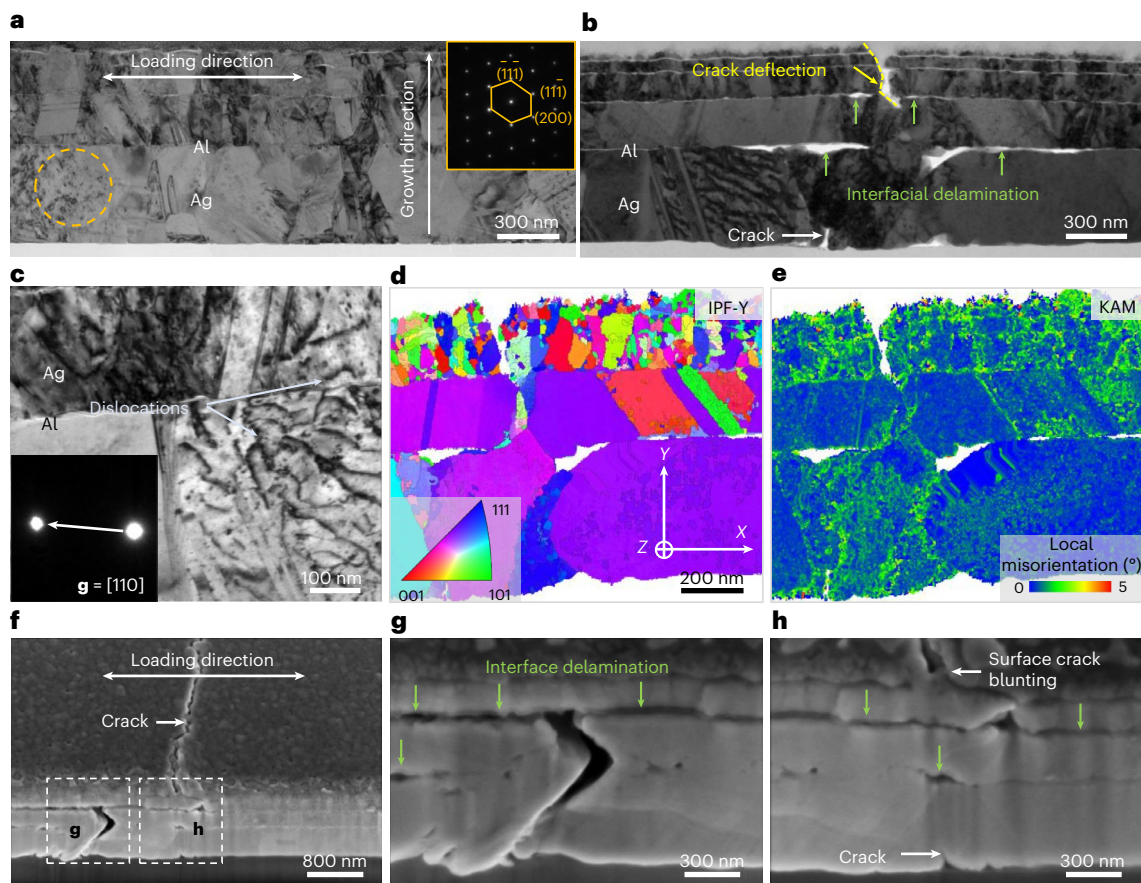


Fig. 3 | Fatigue damage behaviour of the Ag/Al CGNL film. **a**, Cross-sectional TEM image of the Ag/Al CGNL film after 1×10^6 cycles at a maximum strain of 0.7%, with a selected area electron diffraction pattern taken from the orange-circled region. **b**, Cross-sectional TEM image of Ag/Al CGNL film after 5×10^6 cycles at 0.7% strain. **c**, Cross-sectional TEM two-beam diffraction image under $g = [110]$ of the fatigued Ag/Al interface (5×10^6 cycles at a maximum strain of 0.7%), illustrating pronounced dislocation activities in the vicinity of the Ag/Al

interface. **d**, PED inverse pole figure map taken from the damaged region in **b**. **e**, Kernel average misorientation (KAM) map corresponding to **d**, showing the grain misorientations and pronounced dislocation activities across the Ag/Al interface. **f–h**, Cross-sectional SEM images of fatigue cracks, illustrating the crack initiation and propagation behaviour. **g**, Crack initiation at the Ag/Al interface, and interface delamination/crack deflection at the Ag/Al interface. **h**, Crack initiation at the surface and crack blunting.

can be quantitatively evaluated by using linear fracture mechanics⁴⁸. Crack deflection occurs preferentially when the following condition is satisfied⁴⁸:

$$G_D/G_P > \Gamma_{Ag/Al}/\Gamma_{Ag} \quad (1)$$

where G_D and G_P are the energy release rates for a deflected crack and a penetrating crack, respectively, and $\Gamma_{Ag/Al}$ and Γ_{Ag} are the Ag/Al interface toughness and mode I toughness of the Ag layer, respectively. We estimated the mode I toughness experimentally by an approach based on the synchrotron XRD technique (Supplementary Note 5). G_D/G_P is greater than $\Gamma_{Ag/Al}/\Gamma_{Ag}$ when the layer thickness exceeds 60 nm (Supplementary Note 6). It is expected that crack deflection is energetically favourable at the Ag/Al interface for thicker Ag layers, consistent with experimental observations.

Interface delamination and crack deflection are prevalent in CGNL films under both high-cycle (Supplementary Fig. 28) and low-cycle fatigue (Supplementary Fig. 29a). The deflected crack length along the Ag/Al interface in CGNL films can reach several micrometres, almost an order of magnitude larger than the mode I crack length. The crack deflection of the CGNL film is more prominent compared with the Ag and HNL counterparts under the same loading conditions (Supplementary Figs. 20 and 28–30). Activation of such interface delamination is determined by the relatively weak adhesion of the Ag/Al interface³², coupled with the

high-multiaxial-stress state ahead of the propagating crack tip in gradient-structured materials⁴⁹. We used finite-element analysis to compare the stress state near heterophase interfaces and crack propagation behaviour between gradient nanolayered and HNL Ag/Al films. When a crack approaches the interface, the gradient nanolayered film develops much higher stress triaxiality near the interface compared with its homogeneous counterpart (Supplementary Table 4), which drives interfacial delamination and facilitates crack deflection. A pronounced contrast in crack propagation behaviour is revealed by the simulations (Supplementary Figs. 31 and 32). The gradient architecture induces substantial crack deflection, whereas the homogeneous film displays straight-through crack propagation with minimal deflection. These computational predictions correlate well with experimental observations.

Fatigue performance of CGNL films

The electrical degradation is governed by fatigue damage mechanisms including surface extrusions, interface delamination, through-thickness cracks (normal to the film plane) and transverse cracks (propagating along the specimen width) (Supplementary Fig. 33). The minimal electrical degradation in Ag/Al CGNL films can be attributed to two factors: (1) the presence of limited short transverse cracks that partially disrupt lateral conduction pathways and (2) the occurrence of interface delamination that moderately affects out-of-plane conduction pathways.

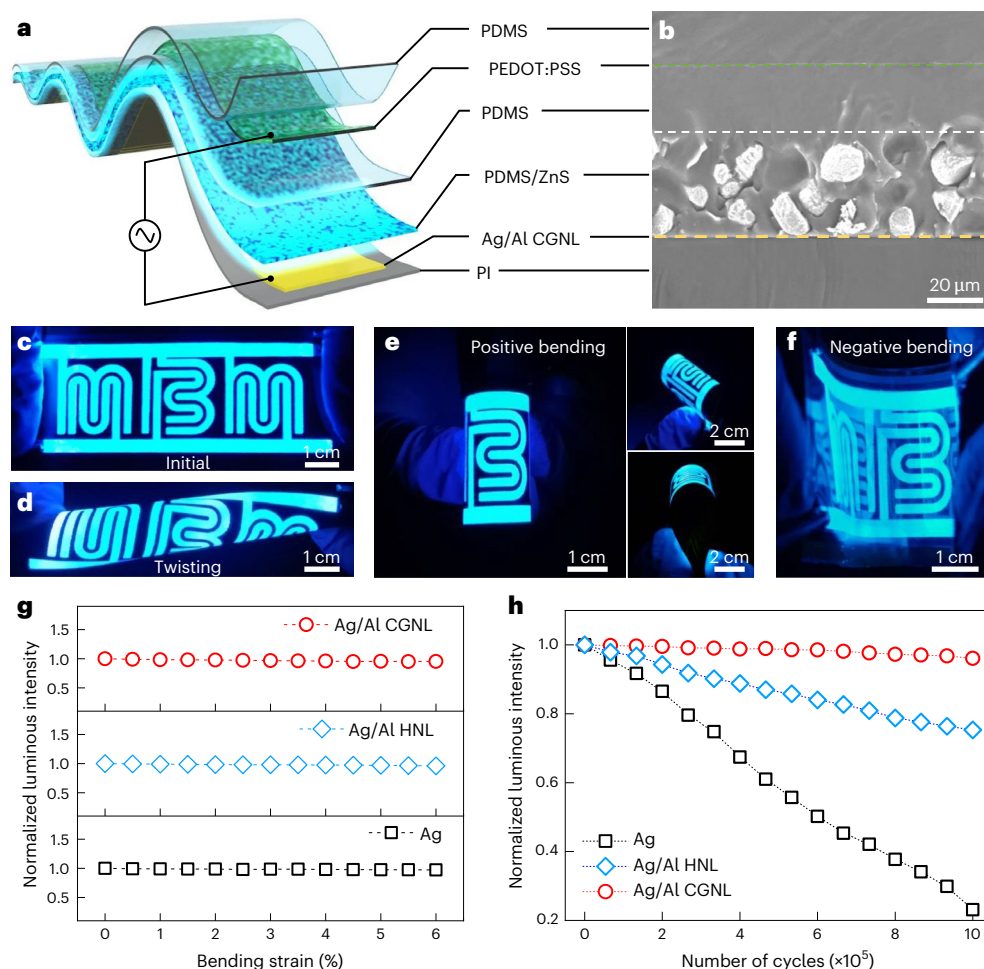


Fig. 4 | Flexible EL device integrated with the fatigue-resistant Ag/Al CGNL film. **a**, Schematic for a flexible EL device, consisting of a package layer (topmost), top electrode, insulating layer, emitting layer, bottom electrode and flexible substrate. **b**, Cross-sectional SEM image of the EL device. **c**, Functionality of a flexible EL device integrated with an Ag/Al CGNL film. **d–f**, Functionality

of the flexible EL device under twisting (**d**), positive bending (**e**) and negative bending (**f**). **g**, Normalized luminous power of the flexible EL device as a function of bending strain for Ag, Ag/Al HNL and Ag/Al CGNL films. **h**, Normalized luminous power versus number of bending cycles for Ag, Ag/Al HNL and Ag/Al CGNL films.

Furthermore, we systematically investigated the influence of multiple factors on the fatigue performances of Ag/Al CGNL films, including structural gradient (Supplementary Fig. 34a1–a3), gradient order (Supplementary Fig. 34b1–b3), coverlay, thermal annealing, substrate thickness and substrate stiffness. First, the fatigue resistance initially increases and then decreases as the structural gradient increases (Supplementary Fig. 35a). Optimal fatigue resistance across a broad strain range (0.7%–5%) is achieved with an intermediate structural gradient (Supplementary Fig. 36). Second, the positive-gradient samples can endure more fatigue cycles than symmetric-gradient and negative-gradient ones (Supplementary Fig. 35b). The dependence of fatigue tolerance on structural gradient and gradient order can be rationalized in terms of crack initiation and crack propagation (Supplementary Note 7 and Supplementary Figs. 37 and 38). Third, coverlay application enhances fatigue resistance, which arises from the mechanical constraint imposed by the coverlay, analogous to that of the polymer substrate. It also reduces the performance disparity across different gradient orders compared with unencapsulated films (Supplementary Note 8 and Supplementary Fig. 39). Moreover, Ag/Al CGNL films exhibit superior thermal stability compared with pure Ag, maintaining unchanged room-temperature fatigue resistance at 373 K and >90% at 473 K (Supplementary Note 9 and Supplementary Figs. 40–42). Finally, fatigue performance generally degrades with either decreasing substrate thickness or stiffness

(Supplementary Figs. 43 and 44), which can be explained by the diminished mechanical constraint from polymer substrates on metal films (Supplementary Note 10).

Our design based on coherent Ag/Al interfaces endows the CGNL films with high electrical conductivity, which is essential for application as flexible conductors. The conductivity of our Ag/Al CGNL film is $3.1 \times 10^7 \text{ S m}^{-1}$ at room temperature (293 K), better than that of Ag films ($2.9 \times 10^7 \text{ S m}^{-1}$). We attribute the enhanced conductivity of the Ag/Al CGNL film to its distinct misorientation angle distribution: a higher frequency of low-angle (0 – 10°) and 55 – 60° boundaries and reduced frequency of intermediate-angle (10 – 55°) boundaries compared with the pure Ag film (Supplementary Note 11 and Supplementary Figs. 45 and 46). Additionally, our strategy of constructing CGNL architectures for fatigue-resistant conductors demonstrates broad applicability beyond the specific Ag/Al system. The design is fundamentally transferable to other conductive material systems such as Cu-, Au- and Al-based thin films (Supplementary Note 12 and Supplementary Figs. 47–52). Importantly, the CGNL design that actively modulates elastoplastic behaviour is well compatible with conventional structural approaches (such as serpentine, buckle or wavy architectures) that confine metals to elastic deformation regimes. Combinations of material and structural design may produce synergistic effects, enabling both greater stretchability and improved long-term durability in flexible electronic devices.

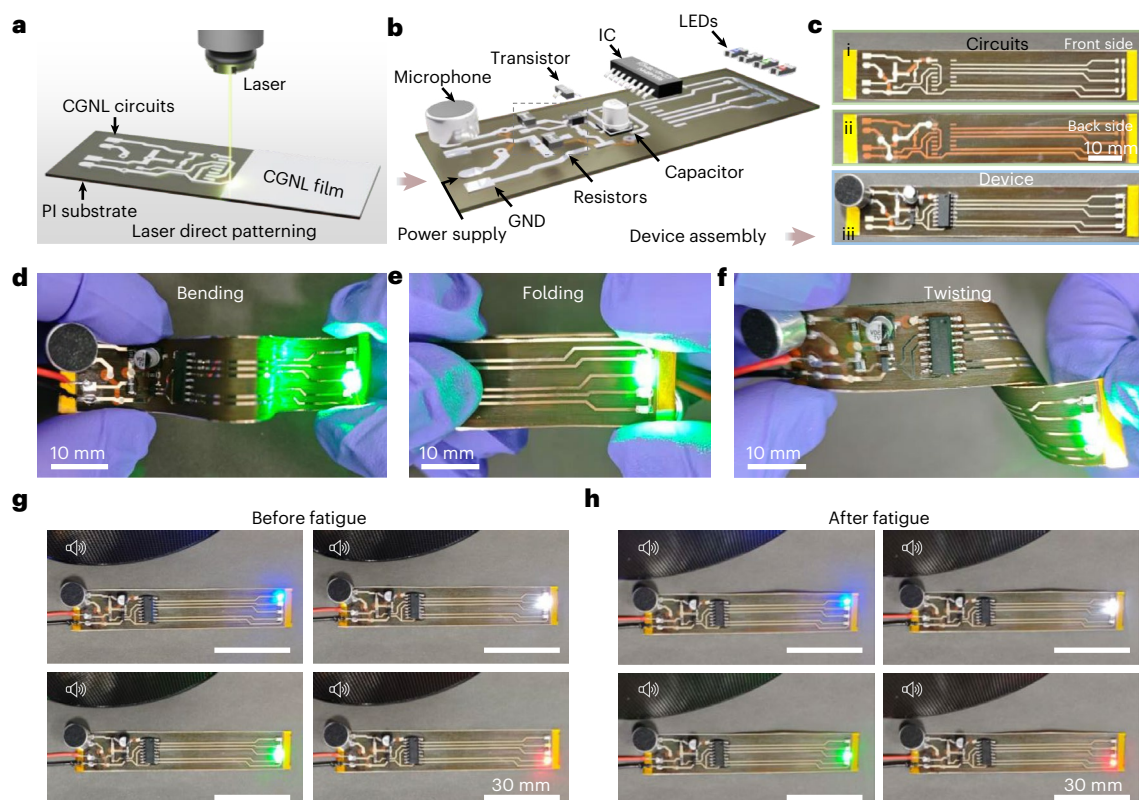


Fig. 5 | Ag/AI CGNL-based flexible functional circuit. **a**, Schematic illustration of laser direct patterning process for CGNL-based flexible circuits. **b**, Exploded schematic of a typical four-colour music-rhythm LED flash circuit. The key components include a CD4017 integrated circuit (IC), a transistor, a capacitor, a microphone, resistors and four-colour LEDs. GND, ground. **c**, Digital images of

the Ag/AI CGNL-based circuit: (i) front view, (ii) back view and (iii) fully assembled device. **d–f**, Flexible functional circuit under bending (**d**), folding (**e**) and twisting states (**f**). **g,h**, Visual display of the LED flash circuit synchronized with music rhythm based on Ag/AI CGNL electrodes, before (**g**) and after fatigue testing (**h**), demonstrating the device's functional stability.

Cycle-insensitive electrical functionalities

To validate the potential of CGNL films as flexible electrodes, we fabricated a model electroluminescent (EL) device as a model system consisting of a bottom opaque electrode (Ag/AI CGNL film), an emitter (ZnS:Cu), an insulating layer (polydimethylsiloxane, PDMS), a top transparent electrode (poly(3,4-ethylenedioxythiophene):poly(styrene sulfonate) (PEDOT:PSS)) and a transparent package layer (PDMS) (Fig. 4a,b). The EL device, reading 'MBM', performed steadily under various deformations, including twisting, positive bending and negative bending (Fig. 4c–f). As a control, we used Ag and Ag/AI HNL as bottom opaque electrodes in the same configuration. Luminous power measurements show that no major change occurs in either EL device based on the three electrodes during bending to 6% strain (Fig. 4g). In contrast to static bending, the luminous power gradually decreases under cyclic bending for Ag and Ag/AI HNL electrode-based devices (Fig. 4h). In contrast, the Ag/AI CGNL electrode-based device exhibits negligible performance degradation after 10^6 cycles at a maximum bending strain of 0.7%. Microstructural analysis confirms that this durability originates from the superior fatigue resistance of the CGNL electrode (Supplementary Fig. 53).

To demonstrate the viability of CGNL films as flexible interconnects, we designed a flexible four-colour musical-rhythm light-emitting diode (LED) synchronization system employing a laser-patterned CGNL film-based circuit (Fig. 5a–c and Supplementary Fig. 54). The integrated system combines commercial integrated circuit chips with standard electronic components including resistors, capacitors and LEDs. The system captures audio signals from a microphone and uses them to drive multistage transistor amplification, achieving rhythm-synchronized switching in the LED array via a CD4017 decade counter. The flexible circuitry maintains full

functionality under various mechanical deformations including bending (radius < 2 mm), 180° folding and 360° twisting (Fig. 5d–f). After 15,000 loading cycles at a maximum strain of 1.2%, Ag film-based circuits exhibit substantial performance degradation (67% luminance reduction, Supplementary Fig. 55), while CGNL-based counterparts maintain stable EL output (<5% variation) (Fig. 5g,h). This superior durability stems from CGNL film's resistance to fatigue-induced cracking, a common failure mechanism observed in Ag film substrates through SEM analysis (Supplementary Fig. 56).

To evaluate the bioelectronic functionalities, we record in real time *in vivo* electromyographic (EMG) signals with a two-channel Ag/AI CGNL electrode based on a rat model. The Ag/AI CGNL electrode was wrapped conformally around the peroneal nerve and muscle (Fig. 6a,b). For comparison, Ag and Ag/AI HNL electrodes were employed in the same configuration. All three electrodes can successfully record subcutaneous compound muscle action potential from the peroneus longus muscle in a rat leg (Fig. 6c). To imitate bioelectronic functionality during long-term mechanical deformation and electrical stimulation, we treated the three electrodes with repetitive bending and electrical stimulation, respectively. After bending deformation for 10^6 cycles at a maximum strain of 0.7%, the Ag/AI CGNL electrodes can still stably detect continuous signals, comparable to those recorded by the undeformed ones, while apparent noise appears in signals collected by Ag electrodes (Fig. 6d). The signal-to-noise ratio (SNR) value of Ag/AI CGNL electrodes is -42 dB, much higher than those of Ag/AI HNL (-26 dB) and Ag (-12 dB) electrodes (Fig. 6f). After electrical stimulation for 3,000 cycles at 200 mA, obvious signal fluctuation and baseline shift are detected in EMG signals recorded by Ag electrodes (Fig. 6e). In contrast, Ag/AI CGNL electrodes still maintain good-quality signal recording with higher SNR and lower baseline noise (Fig. 6f,g).

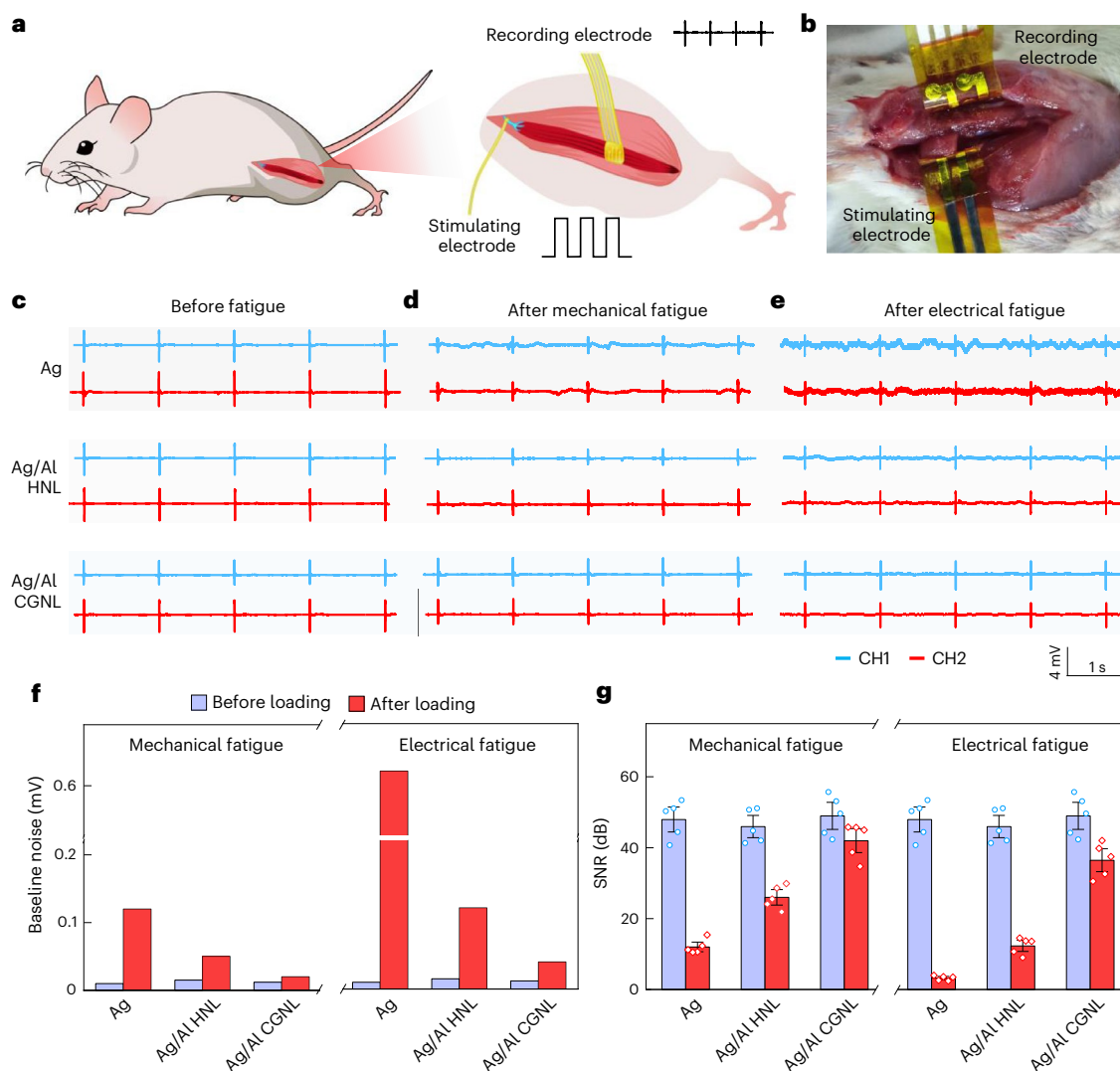


Fig. 6 | In vivo recording and stimulation via the Ag/Al CGNL bioelectrodes.

a, Schematics of simultaneous compound muscle action potential stimulation and recording via bioelectrodes. In this configuration, both stimulation and recording electrodes are CGNL electrodes. **b**, Digital image of stimulation and recording of simultaneous compound muscle action potential. **c**, Representative EMG signals recorded by a two-channel (CH1 and CH2)

electrode based on as-fabricated Ag, Ag/Al HNL and Ag/Al CGNL films, respectively. **d**, **e**, EMG signals recorded with Ag film, Ag/Al HNL and Ag/Al CGNL electrodes, respectively, after mechanical fatigue (**d**) and electrical fatigue (**e**). **f**, **g**, Comparison of baseline noise (**f**) and SNR (**g**) of EMG signals collected from the three electrodes after mechanical and electrical fatigue. The error bars are the s.d. of the mean ($n = 5$ peaks).

This can be attributed to the severe fatigue damage (extrusions, grain boundary grooves and cracks) in Ag electrodes induced by repetitive electrical stimulation (Supplementary Fig. 57).

Conclusion

We have reported the development of metal-film-based conductors with high electrical conductivity and fatigue resistance using a CGNL architecture. Our Ag/Al CGNL films exhibit cycle-insensitive conductivity in both the high-cycle regime, maintaining a conductivity of around 10^7 S m^{-1} over 10^7 cycles at strains ranging from 0.7% to 2%, and low-cycle regime, maintaining a conductivity higher than 10^6 S m^{-1} over 10^5 cycles at 5% strain. The high fatigue resistance can be attributed to HDI strengthening, controlled grain coarsening and mitigated interface stress concentration delaying fatigue-crack initiation, and moderate interface adhesion and a gradient-structure-induced multiaxial stress state promoting interface delamination and fatigue-crack deflection. We illustrate the potential of our CGNL films as fatigue-resistant components in robust electrodes, interconnects and bioelectronic interfaces,

which all require long-term functional stability. Our CGNL design is independent of existing strategies for improving fatigue resistance (such as interlayers and strain-relief structures) and is compatible with silicon-based microelectronic manufacturing technology.

Methods

Film preparation

Ag/Al CGNL films were prepared on PI substrates by direct current magnetron sputtering at room temperature. The Ag layers with thickness gradient containing nominal thickness of 30/60/150/250/500 nm layers from top (surface) to bottom (substrate) and Al layers with constant thickness of ~4 nm were deposited alternately using high-purity Ag (99.999%) and Al (99.95%) targets. Before deposition, the PI substrates were ultrasonically cleaned using acetone and then rinsed in deionized water. The base vacuum before sputtering was 1×10^{-4} Pa, and the Ar pressure during sputtering was 0.5 Pa. The deposition rate of Ag and Al was -0.45 nm s^{-1} and -0.2 nm s^{-1} , respectively. Additionally, Ag/Al CGNL films with the same gradient order but different

thickness gradients (30/60/150/250/500 nm, 10/20/30/40/890 nm and 140/170/200/230/250 nm) and the same thickness gradient but different gradient orders (30/60/150/250/500 nm (positive gradient), 500/250/150/60/30 nm (negative gradient) and 30/215/500/215/30 nm (symmetric gradient)) were prepared (Supplementary Fig. 34). As controls, HNL Ag/Al films (200-nm sublayers) and single-layer Ag films were prepared, each with a total thickness of 1,000 nm. For direct comparison of interface effects, we designed semicoherent Ag/Mo gradient nanolayered films with identical architectural configuration (30/60/150/250/500 nm sublayers) to the Ag/Al system (details in Supplementary Note 4). To further demonstrate the universality of our fatigue-resistant design strategy, we fabricated CGNL films using two additional material systems: Ag/Au and Cu/Ni (Supplementary Note 12).

Microstructure observations

Cross-sectional TEM samples were prepared with a focused ion beam (FIB) instrument (Thermo Scientific Helios 5 UX) by using a lift-out methodology. Before FIB milling, thin C and Pt protective layers were coated on the contact areas. The final FIB milling voltage/current was set as 2 kV/10 pA, which was sufficiently small to avoid potential irradiation damage and remove residual Ga⁺ and Pt. Planar TEM specimens were prepared by mechanical milling, followed by ion milling (Gatan Pips model 691) with a final beam voltage of 3.5 kV and milling angle of $\pm 4^\circ$. Cross-sectional microstructures before and after cyclic loading were characterized by using a field emission TEM (JEOL-2100F) operating at an accelerating voltage of 200 kV. Atomic-resolution HAADF-STEM imaging and EDS were carried out by using an aberration-corrected scanning transmission electron microscope (JEM-ARM200F) operated at 200 kV, with a probe convergence semi-angle of 21 mrad. Characterizations of grain orientations in planar view were carried out using the on-axis TKD technique, using a Zeiss Gemini 500 scanning electron microscope equipped with an Oxford Instruments Channel 5 electron backscatter diffraction system. The acceleration voltage and step size for TKD are 30 kV and 20 nm, respectively. The planar TEM specimens can also be used for TKD analyses. Orientation mapping and grain-boundary characterization in cross-sectional view were obtained with the PED technique using NanoMEGAS (Brussels, Belgium) hardware and a NanoMEGAS ASTARTM system. A precession angle of 0.6° and a step size of 3 nm were used. The phase structures were determined with XRD using a Bragg–Brentano diffractometer in $30\text{--}90^\circ$ configuration (Bruker D8 ADVANCE) with Cu K α radiation. The surface and cross-sectional damage morphologies before and after cyclic loading were characterized by using a scanning electron microscope (Zeiss Sigma 300). The samples for cross-sectional morphology observations were prepared with a FIB instrument (Thermo Scientific Helios 5 UX). The three-dimensional surface damages were characterized using atomic force microscopy (Bruker Dimension Icon).

Mechanical- and electrical-property testing

Mechanical-property measurements, including monotonic tension, monotonic bending, tension–tension fatigue and bending–bending fatigue tests, were conducted by using a Microforce Test System (MTS Tytron 250). During mechanical testing, the electrical resistance was measured in real time using a four-point measurement configuration with a digital source meter (Keithley 2601A). In this set-up, a constant current is applied, while the voltage is simultaneously measured. All electrical contacts are integrated into the mechanical grips (Supplementary Fig. 58). Secure mechanical clamping and stable electrical contact are maintained throughout testing to ensure measurement accuracy. Each custom-designed clamp comprises a base part made of stainless steel and a top part made of stainless steel coated with insulation. A polytetrafluoroethylene buffer interlayer is implemented between the steel plates and the specimen to homogenize stress distribution at the clamping interfaces and mitigate localized stress concentration. The sandwich structure (top plate, buffer layer, specimen,

buffer layer, base plate) is secured using two screws to ensure uniform pressure distribution (Supplementary Fig. 58). Post mortem SEM analysis of the fatigued specimen demonstrated negligible influence of clamp-induced stress perturbations (Supplementary Fig. 59). The specimens have a gauge section of 30 mm in length and 5 mm in width. Three individual experiments were performed for each experimental condition. Tension tests were performed at a constant strain rate of $1 \times 10^{-3} \text{ s}^{-1}$. Tension–tension fatigue tests were performed under constant total strain control. The maximum strain varies from 0.7% to 5% and a minimum strain of 0.2% was applied to prevent sample buckling. A sinusoidal waveform with frequencies of 5 Hz and 1 Hz was used for 0.7–2.0% and 5% strain. Note that the maximum strains are much lower than the cracking onset strains of the films (Supplementary Figs. 60 and 61). Monotonic bending and cyclic bending–bending tests were conducted by moving one clamp closer to the other at a constant speed of 0.03 mm s^{-1} . Bending strain can be determined from the analytical formula $\varepsilon_b \approx t/2r$ (Supplementary Fig. 62), where t is the total thickness of the film/substrate system and r is the radius of curvature, which can be obtained by image analysis (ImageJ).

In situ uniaxial tension tests combined with synchrotron XRD were carried out at the 16B1 beamline, Shanghai Synchrotron Radiation Facility, China. The beam wavelength was 0.1239 nm and the spot size at the sample was $0.39 \times 0.48 \text{ mm}^2$. Two-dimensional diffraction patterns were collected using a Mar165 CCD (charge-coupled device) detector. The distances from samples to detectors are roughly 54 mm. All samples were loaded stepwise to a strain of 6% under a constant strain rate of 10^{-3} s^{-1} . The tensile stage was mounted in the centre of the goniometer, with the incident X-ray beam oriented perpendicular to the sample surface. Experiments were performed using a beam energy of 10 keV, and exposure time of 30 s. The elastic lattice strains were deduced from the elliptical distortions of the Ag (111) Debye–Scherrer rings. The film stress can be obtained from the measured lattice strains using the $\sin^2\psi$ method⁵⁰. The yield strength is defined as an offset plastic strain of 0.2% from the strain–stress curves⁵¹.

Electrical-fatigue testing

During long-term cyclic electrical stimulation, repetitive volumetric expansion/shrinkage of the film/substrate system can induce fatigue damage, deteriorating the recording and stimulating efficacy^{52,53}. In electrical-fatigue tests, the specimens were well fixed using flat electrically conductive clips (Supplementary Fig. 63). An infrared thermal imager (Fluke 971) was utilized to monitor the variation of sample temperature (Supplementary Fig. 64). The stimulation pulse was performed using an intelligent dual-pulse power supply (GKP-FB4-30 A/15 V, Shenzhen SC Electronic Technology Co. Ltd) in a monophasic pulse mode. The current is 200 mA, and the width is 2 ms. The stimulation period was set as 6 s to allow the specimen to completely cool (Supplementary Fig. 64). After cyclic electrical stimulation, the specimens were used as bioelectrodes to evaluate the long-term performance of electrophysiological stimulating and recording.

Fabrication of EL device

A flexible EL device consisting of an emitting layer (ZnS:Cu/PDMS composite), insulating layer (PDMS, Sylgard184, Dow Corning), top electrode (poly(3,4-ethylenedioxythiophene) polystyrene sulfonate, PEDOT:PSS) and package layer (PDMS) was prepared using layer-by-layer spin-coating on the Ag/Al CGNL film-coated PI substrate. For spin-coating, the PI substrate was first fixed on a 4-in Si wafer. The Cu-doped ZnS particles were dispersed in PDMS prepolymer and cross-linker with a weight ratio of 15:10:1 and then the mixture was well stirred at a speed of 2,000 rpm for 20 min. The spin-coating speeds of the emitting, insulating and package layers are set as 2,000, 3,000 and 3,000 rpm, corresponding to the thicknesses of ~ 50 , ~ 20 and $\sim 20 \mu\text{m}$, respectively. The top transparent electrode was prepared using spin-coating at 2,000 rpm on the cured insulating PDMS layer.

To improve the hydrophilicity of the PDMS layer, a 5 min O₂ plasma treatment was performed before spin-coating. After each spin-coating, the assembly was cured in an 80 °C oven for 2 h. The EL device was powered by a direct-current 3 V adaptor inverter (B-1, Shenzhen Oubait Technology Co. Ltd) generating alternating current with a voltage amplitude of 110 V and a frequency of 200 Hz. The luminance of the EL device during monotonic bending and bending–bending fatigue tests was measured using an illumination photometer (Fluke 941).

Fabrication of Ag/Al CGNL-based functional circuit

Functional circuits were fabricated on 125-µm-thick PI substrates coated with Ag/Al CGNL films on both sides using magnetron sputtering. Both the top and bottom circuit layers underwent single-pass laser exposure at an average power of 5 W using a 355-nm ultraviolet laser system (RC1001C, Sino-Galvo (Jiangsu) Technology Co. Ltd). Vertical interconnect accesses connecting these two layers were subsequently formed via laser ablation. These vertical interconnect accesses were then filled with commercial solder paste. Finally, electronic components were placed onto the solder-paste-coated pads and soldered to the multilayered circuit board using a hot-air reflow system.

In vivo EMG recording

Two-channel electrodes for in vivo EMG recording were prepared on 25-µm-thick PI substrates by magnetron sputtering (ULVACUL-VAC, ACS-4000) with a mask. All electrodes were fabricated in an aseptic manner and were further disinfected under ultraviolet light for 3 h. Three-month-old adult female Sprague Dawley rats (~250 g) were used to evaluate the in vivo EMG recording functionalities of our specimens before and after fatigue. Before the experiment, the test rats were anaesthetized by injecting sodium pentobarbitone (density 2%, dose 0.3 ml per 100 g) into the intraperitoneal space. During in vivo experiment, the rats were comfortably placed over a heating pad with a constant temperature of 37 °C on the operating table. EMG signals were collected with a data acquisition and analysis system (BL-422I, TAIMENG Software). The SNR can be calculated from $SNR = 20 \log(S_{RMS}/N_{RMS})$, where S_{RMS} is the root mean square of the signal amplitude recorded when the rat was stimulated and N_{RMS} is the root mean square of the noise amplitude recorded in the rest state. In vivo experiments were performed in Xi'an Jiaotong University, and were approved by and adhered to the guidelines of its institutional animal care and use committee (No. 2019–773).

Computational modelling

Molecular dynamics simulation. Molecular dynamics simulations were performed using the Large-Scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) package to investigate interface-related fatigue damage behaviour. We used the empirical embedded atom method potential developed by Zhou et al.⁵⁴ to describe the interatomic interaction for both Ag–Al and Ag–Mo systems. The Ag/Al/Ag multilayer structure is oriented as x [112], y [110] and z [111] for Ag and Al (Supplementary Fig. 26a). The Ag/Mo/Ag multilayer structure is oriented as x [112], y [110] and z [111] for Ag and x [211], y [111] and z [011] for Mo. The orientation of the interface for Ag/Mo is followed using the Kurdjumov–Sachs orientation relationship of $[110]_{FCC}/[111]_{BCC}$, as shown in Supplementary Fig. 26b. The sample has dimensions of 32 nm × 15 nm × 46 nm with the thickness of Ag along the z direction ~20 nm and Mo (Al) along the z direction ~6 nm. Periodic boundary conditions are applied in all dimensions. Cyclic loading simulations were performed parallel to the y axis within a 2% strain range (see Supplementary Fig. 26c), under the NVT ensemble and at a constant strain rate of $1 \times 10^9 \text{ s}^{-1}$. This strain range was selected to ensure that the Ag/interlayer/Ag multilayer system remained in the elastic deformation stage after the first loading cycle, consistent with experimental findings. The temperature was kept at 300 K using the Nosé–Hoover

thermostat^{55,56}. All simulations were carried out using the LAMMPS code⁵⁷. The atomic configurations were displayed using OVITO⁵⁸.

Finite-element simulations. Finite-element simulations were performed to investigate stress distributions and crack propagation in Ag/Al gradient nanolayered samples. The gradient model comprised five layers (H1–H5) with the following thicknesses: H1, 30 nm; H2, 60 nm; H3, 150 nm; H4, 250 nm; H5, 500 nm. Model parameters were calibrated using experimental data. Given the Al interlayer's ultralow thickness relative to the Ag layers and its predominant function in interfacial adhesion rather than mechanical load-bearing, we computationally modelled the Al interlayer as a two-dimensional interface. This simplification effectively reduces computational complexity while retaining the accuracy of crack propagation predictions. The interfacial strength is defined as⁵⁹

$$\sigma_{ad} = \sqrt{\frac{2\gamma_{Ag/Al}E}{c} \left(1 - \frac{c}{h}\right)} \quad (2)$$

where $\gamma_{Ag/Al}$ is the adhesive energy of the Ag/Al interface, E is the elastic modulus of Ag, c is the crack length and h is the thickness of the Ag/Al film.

The conventional mechanism-based strain gradient theory⁶⁰ was applied in analysing crack propagation within gradient films (Supplementary Note 13). The cohesive zone model was integrated to investigate the crack propagation behaviour observed across different layers. Ag grain-boundary strength was assumed to be consistent. Microcracks with lengths comparable to the local layer thickness were pre-introduced at two distinct locations (the top surface layer and the bottom thick layer of the gradient films, respectively) to investigate the influence of crack initiation sites on propagation behaviour. Upon subsequent loading, the pre-existing microcrack propagates and reaches the interface. This simulation design with pre-introduced cracks can capture the crack-tip stress distribution near the interface and characterize fatigue-crack propagation behaviour. All simulations were conducted using Abaqus software.

Data availability

The data that support the findings of this study are available from the corresponding authors on reasonable request.

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Author contributions

K.W. and J.S. supervised the project and generated the design concept. Y.X. conducted the majority of experiments. T.Z. helped to fabricate the EL device and data analysis. B.L. and K.C. contributed to the design of the in vivo experiments. B.C. conducted the molecular dynamic simulation. Y.L. and Q.Z. performed the finite-element analysis. Y.X., K.W., Y.W., J.Z., G.L. and J.S. analysed the data and discussed the results. Y.X. and K.W. wrote the paper. All authors reviewed and contributed to the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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